

Shahid Anwar

Data Scientist | Machine Learning Engineer | AI Researcher

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PROFESSIONAL SUMMARY

Data scientist and machine learning engineer with a Master's in Artificial Intelligence from the University of Stirling, an MS in Computer Science and a peer-reviewed IEEE publication in ensemble learning. Experienced in designing reproducible model evaluation pipelines across supervised, unsupervised and deep learning in Python (scikit-learn, PyTorch, TensorFlow). Skilled at identifying data leakage, overfitting and class-imbalance issues, selecting sound cross-validation strategies, and communicating complex algorithmic and statistical results clearly in writing, supported by two years of university lecturing and code assessment.

CORE COMPETENCIES

Machine Learning & Deep Learning: Supervised and unsupervised learning, ensemble learning, anomaly detection, CNNs, Vision Transformers, computer vision, NLP, explainable AI (Grad-CAM)

Model Evaluation & Statistics: Cross-validation strategies, stratified splits, hyperparameter optimisation, imbalanced-data handling, leakage-aware testing, statistical inference, dimensionality reduction, macro F1, ROC analysis, quadratic weighted kappa

Programming & Data: Python, Pandas, NumPy, scikit-learn, PyTorch, TensorFlow, SQL, FastAPI, Kotlin, Java, Flutter, Dart, PHP, JavaScript

AI Systems & Workflow: LLM applications, retrieval-augmented generation (RAG), mixed-precision training, OneCycleLR scheduling, early stopping, reproducible experiment design

Communication: Technical and academic writing, code review, structured feedback, teaching and documentation

SELECTED DATA SCIENCE AND RESEARCH PROJECTS

Prostate Cancer Histopathology Classification - PANDA Dataset

Computer Vision | CNN | ViT-B/16 | GCViT-Tiny | PyTorch

- Built a reproducible image-classification pipeline on 10,616 histopathology images using stratified train, validation and test splits to prevent data leakage.
- Benchmarked a CNN, Vision Transformer (ViT-B/16) and GCViT-Tiny under controlled experimental settings.
- Mitigated class imbalance and overfitting with class-weighted cross-entropy, label smoothing, OneCycleLR scheduling and early stopping; used mixed-precision training for GPU efficiency.
- Achieved test accuracy of 0.7872 (CNN) and 0.7994 (ViT-B/16); evaluated multiclass performance with macro F1, confusion matrices and quadratic weighted kappa.
- Applied Grad-CAM to audit which image regions drove predictions, validating model reasoning and interpretability.

Adaptive Anomaly-Based Intrusion and Ransomware Detection Framework

Unsupervised Learning | Ensemble Learning | CICIDS2017 | UNSW-NB15 | scikit-learn

- Designed a lightweight unsupervised ensemble combining Isolation Forest, Local Outlier Factor and One-Class SVM with adaptive thresholding.
- Developed end-to-end preprocessing, training and evaluation workflows across network-intrusion and ransomware datasets.
- Recorded 0.99924 accuracy with random forest on CICIDS2017 and 0.9958 with logistic regression on UNSW-NB15, and scrutinised near-perfect scores for leakage and dataset bias.
- Evaluated true generalisation with leave-family-out testing, achieving 0.8909 macro accuracy across unseen ransomware families.
- Assessed robustness and operational trade-offs using classification reports, ROC analysis, confusion matrices and comparative experiments.

Adaptive Ensemble Learning for Anomaly-Based DDoS Detection

Peer-Reviewed Publication | IEEE WINCOM 2023

- Developed and evaluated an ensemble-based approach for detecting anomalous DDoS network traffic.
- Contributed to experimental design, model comparison and written research communication for peer-reviewed publication.

PROFESSIONAL EXPERIENCE

Lecturer - Visiting Faculty

GC Women University Sialkot, Pakistan | Feb 2023 - Jan 2025

- Delivered undergraduate teaching in Web Development, Android Mobile Application Development and Operating Systems.

- Reviewed and graded student code and projects, identifying logical, syntax and implementation errors and providing structured, actionable feedback.
- Explained complex technical and algorithmic concepts clearly through written materials, applied examples and practical demonstrations.
- Prepared learning material and assessments aligned with module learning outcomes.

Assistant Computer Programmer

GC Women University Sialkot, Pakistan | Apr 2022 - Jan 2025

- Delivered front-line technical support to faculty and students, independently diagnosing and resolving software, account and connectivity issues.
- Configured eduroam and supported Microsoft 365, OneDrive, learning-management systems and digital catalogue services.
- Maintained university websites, OPAC/library web services and in-house applications, coordinating updates with academic teams.
- Identified and implemented practical improvements to IT service workflows and helpdesk operations.

Senior Software Engineer

eddipanon.com, Pakistan | May 2020 - Jan 2022

- Developed real-time chat functionality using Firebase for data storage, synchronisation and retrieval.
- Implemented secure user authentication and contributed to maintainable, tested application architecture.
- Improved cross-platform compatibility, contributing to a reported 92% increase in user engagement.
- Translated product requirements into tested features in collaboration with stakeholders.

Computer Operator and Teacher

Little Angels Home School, Sialkot Cantt, Pakistan | Jun 2018 - Oct 2018

- Maintained school data systems and delivered foundational computing instruction.

EDUCATION

Master's in Artificial Intelligence | University of Stirling, Scotland, United Kingdom

MS in Computer Science | University of Management and Technology, Sialkot, Pakistan

BS Software Engineering | University of Management and Technology, Sialkot Campus, Pakistan

PUBLICATION

An Adaptive Ensemble Learning Approach for Anomaly-Based DDoS Detection. IEEE WINCOM 2023. DOI: 10.1109/WINCOM59760.2023.10323025

ADDITIONAL INFORMATION

Certification: Python Bootcamp - Udemy

Languages: English - Fluent | Urdu - Native

Availability: Remote, flexible schedule; 10-40 hours per week

Professional Interests: Machine learning evaluation, AI model reasoning, data science, cybersecurity analytics